

ERD Exercise 1: Student–Academic Advisor

A college maintains information about its **students** and **academic advisors**.

Each student is uniquely identified by a **StudentID** and has a **Name**.

Each academic advisor is uniquely identified by an **EmployeeID** and has an **AdvisorName**.

Every **student must be assigned to exactly one academic advisor**. Each **academic advisor must advise one or more students**.

Using **Crow’s Foot notation**, draw an ERD that includes the appropriate:

1. Entities and attributes
 2. Primary keys
 3. Foreign key
 4. Cardinality and ordinality for the relationship between Student and Academic Advisor
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ERD Exercise 2: Vehicle–Parking Permit

A university issues a **parking permit** to each registered vehicle.

Each vehicle is identified by a **LicensePlate, Make, Model, and Color**.

Each parking permit is uniquely identified by a **PermitID**.

Every **vehicle must have exactly one parking permit**, and every **parking permit must be assigned to exactly one vehicle**. A parking permit cannot be shared by multiple vehicles.

Using **Crow’s Foot notation**, draw an ERD that includes the appropriate entities, attributes, primary keys, foreign key, and the correct cardinality and ordinality between **Vehicle** and **Parking Permit**.

ERD Exercise 3: Author–Book

A publishing company keeps track of its **authors** and the **books** they write.

Each author is uniquely identified by an **AuthorID** and has a **Name** and **Email**. Each book is uniquely identified by a **ISBN** and has a **Title and Edition**.

Each **author must write one or more books**, and each **book must have one or more authors**. Therefore, an author may write many books, and a book may be written by many authors.

Using **Crow's Foot notation**, draw an ERD showing the appropriate entities, attributes, primary keys, and the correct cardinality and ordinality between **Author** and **Book**. For this exercise, you are allowed to have a logical M:N cardinality.

ERD Exercise 4: Resolve an M:N Relationship with an Associative Entity

Use the ERD from Ex 3. Because this creates a **many-to-many (M:N) relationship**, it must be resolved using an associative entity called **Authorship**.

The **Authorship** entity should contain the primary keys from both **Author** and **Book** as foreign keys. These **two** attributes should also form a **composite primary key** for Authorship.

Using **Crow's Foot notation**, redraw the ERD so that the original M:N relationship between Author and Book is replaced with two 1:N relationships through the **Authorship** associative entity. Show the appropriate primary keys, foreign keys, cardinalities, and ordinalities.

ERD Exercise 5: Extend Ex 4 to include Publisher

The publishing company also keeps track of its books.

Each **book must be published by exactly one publisher**, while each **publisher may publish one or more books**.

Each publisher is uniquely identified by its **publisherID** and has a **company name, address, phone, email**.

Using **Crow's Foot notation**, extend the ERD showing the appropriate entities, attributes, primary keys, foreign keys, cardinalities, and ordinalities for **Author, Authorship, Book, and Publisher**.